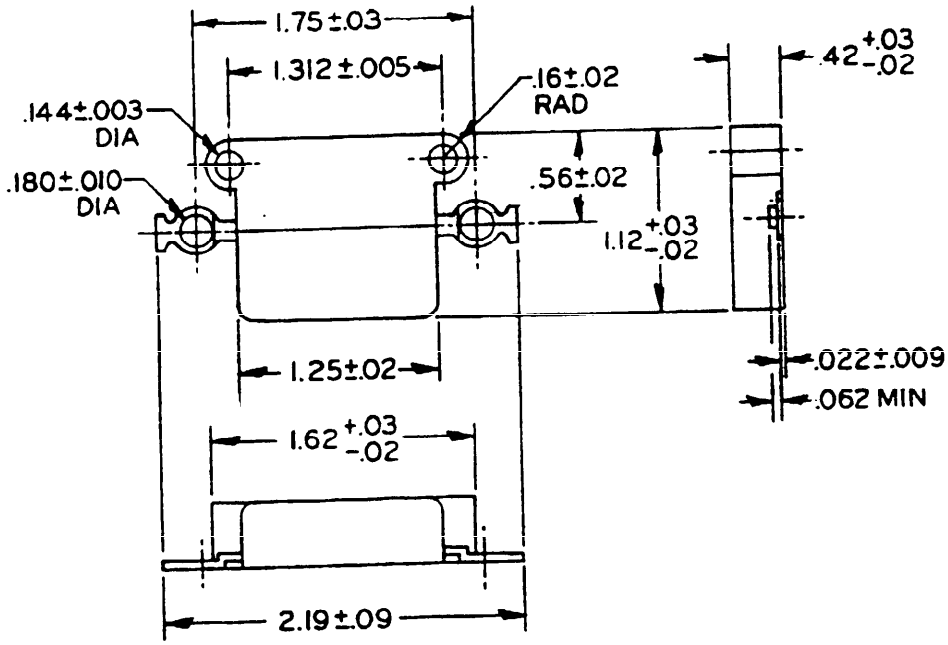
INCH-POUND
MIL-C-5/8G
7 August 1991
SUPERSEDING
MIL-C-5/8F
12 August 1966

MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, MICA DIELECTRIC, STYLE CM50

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-5.



Inches	mm
.003	.08
.005	.13
.009	.23
.010	.25
.02	.51
.022	.56
.03	.76
.062	1.57
.09	2.29
.144	3.66
.16	4.06
.180	4.57
.33	8.38
.56	14.22
1.12	28.45
1.25	31.75
1.312	33.32
1.62	41.15
1.75	44.45
2.19	55.63

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Approximate weight .92 ounce.

(G) FIGURE 1. Dimensions.

(G) denotes changes

FSC 5910

AMSC N/A

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REQUIREMENTS:

(G) Qualification inspection: Qualification inspection is not required.

(G) Dimensions: See figure 1.

Capacitance value: See table I.

Capacitance tolerance: See table I.

DC rated voltage: See table I.

Operating temperature range: See table I.

Symbol	Temperature range
N	-55°C to +85°C
O	-55°C to +125°C
P	-55°C to +150°C

Vibration grade: 3 (10 to 2,000 cps).

Characteristic:

Symbol	Temperature coefficient P/M/°C	Capacitance drift
B	Not specified	Not specified

Dielectric withstanding voltage: Method 301 of MIL-STD-202. 200 percent of dc working voltage for not less than 1 second nor more than 5 seconds.

Barometric pressure: Method 105 of MIL-STD-202, condition B (50,000 feet).
Test potential: 150 percent of rated dc voltage for not less than 1 second nor more than 5 seconds.

Insulation resistance: Method 302 of MIL-STD-202, condition A.

At 25°C - 10,000 megohms minimum.
At 85°C - 4,000 megohms minimum.
At 125°C - 1,500 megohms minimum.
At 150°C - 750 megohms minimum.

Dissipation factor, maximum: See MIL-C-5.

Capacitance: Method 305 of MIL-STD-202.

Vibration: Method 204 of MIL-STD-202, condition D (20 G).

(G) Temperature cycling: Method 107 of MIL-STD-202, condition A.

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Immersion: Method 104 of MIL-STD-202, condition B.

Dielectric withstanding voltage: Meet initial requirement.

Insulation resistance: 30 percent of initial requirement.

Dissipation factor: Not greater than 150 percent of initial value.

Capacitance change: Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

Shock (medium impact): Method 205 of MIL-STD-202, condition C (50 G).

Moisture resistance: Method 106 of MIL-STD-202.

Dielectric withstanding voltage: Meet initial requirement.

Insulation resistance: At least 30 percent of initial requirement.

Dissipation factor: Not greater than 150 percent of initial value.

Capacitance change: Shall not exceed ±4 percent of nominal value or 1 pF, whichever is greater.

Temperature coefficient and capacitance drift: Not specified.

(G) Low temperature storage and life: 48 hours at -55°C +0°C, -2°C.

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TABLE I. Style CM50.

Part or Identifying Number (PIN) 1/ 2/	Capacitance (pF)	DC working voltage (volts)	Operating temperature range	Characteristics available for capacitance tolerance	
				±2% (G)	±5% (J)
CM50BK202--3	2,000	2,500	NOP	B	B
CM50BK222--3	2,200	2,500	NOP	B	B
CM50BK242--3	2,400	2,500	NOP	B	B
CM50BK272--3	2,700	2,500	NOP	B	B
CM50BK302--3	3,000	2,500	NOP	B	B
CM50BK332--3	3,300	2,500	NOP	B	B
CM50BK362--3	3,600	2,500	NOP	B	B
CM50BK392--3	3,900	2,500	NOP	B	B
CM50BK432--3	4,300	2,500	NOP	B	B
CM50BK472--3	4,700	2,500	NOP	B	B
CM50BK512--3	5,100	2,500	NOP	B	B
CM50BG562--3	5,600	1,200	NOP	B	B
CM50BG622--3	6,200	1,200	NOP	B	B
CM50BG682--3	6,800	1,200	NOP	B	B
CM50BG752--3	7,500	1,200	NOP	B	B
CM50BG822--3	8,200	1,200	NOP	B	B
CM50BG912--3	9,100	1,200	NOP	B	B
CM50BG103--3	10,000	1,200	NOP	B	B
CM50BG113--3	11,000	1,200	NOP	B	B
CM50BE123--3	12,000	600	NOP	B	B
CM50BE133--3	13,000	600	NOP	B	B
CM50BE153--3	15,000	600	NOP	B	B
CM50BE163--3	16,000	600	NOP	B	B
CM50BE183--3	18,000	600	NOP	B	B
CM50BE203--3	20,000	600	NOP	B	B
CM50BE243--3	24,000	600	NOP	B	B
CM50BE273--3	27,000	600	NOP	B	B

1/ The term Part or Identifying Number (PIN) is equivalent to the term type designation which was previously used in this specification.

2/ Complete PIN will include additional symbols to indicate capacitance tolerance and temperature range.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17, 99
DLA - ES

User activity:
Army - ME

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5910-1747)

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